

Performance Testing

Date	10 July 2026
Team ID	xxxxxx
Project Name	Rising Waters – Flood Prediction system
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Google Colab, Python
Type of Testing	Performance Testing
Target Module / API	Flood prediction model
Test Environment	Local (Google Colab)
Test Date	14 -07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load flood dataset	1	5	Dataset loaded successfully
2	Data preprocessing	1	10	Data cleaned without errors
3	Model prediction	1	15	Flood prediction generated correctly
4	Performance evaluation	1	10	Accuracy displayed successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Within target limit
3	Throughput (Req/sec)		22 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No errors detected
5	CPU Utilization	< 80%	45%	Pass	CPU usage is normal
6	Memory Utilization	< 80%	52%	Pass	Memory usage is stable

Step 4: Observations & Analysis

Key Findings :

The Flood Prediction System performed successfully during testing. Response time was within the expected range, no critical errors were observed, CPU and memory usage remained stable, and the machine learning model generated accurate flood predictions

Bottlenecks Identified :

No major bottlenecks were identified during testing. Minor delays were observed only during the initial dataset loading phase, but they did not affect the overall prediction performance or system accuracy.

Optimization Steps Taken:

Data preprocessing was optimized by removing missing values and duplicate records. The machine learning model parameters were tuned to improve prediction accuracy.

Unnecessary computations were reduced, resulting in faster execution and efficient

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]

